

# Signorini's Problem

Max Stuthmann

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## 1 Introduction

## 2 Sobolev Spaces

- Integer Order Sobolev Spaces
- Fractional Order Sobolev Spaces
- Trace Operators and Boundary Spaces
- $H(\operatorname{div}; \Omega)$  and Normal Traces

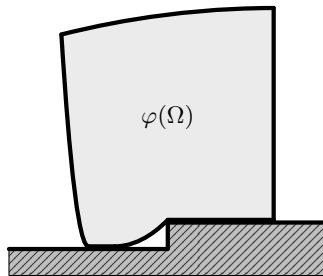
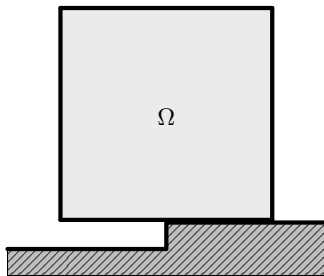
## 3 Signorini's Problem

- Geometric Setup
- Weak Formulations
- Strong Formulations
- Mixed Formulations

## 4 Further Topics

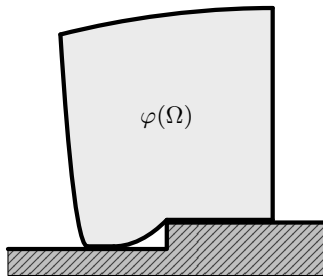
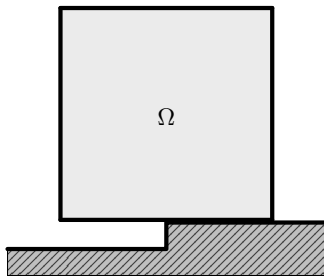
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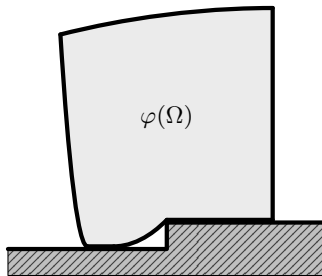
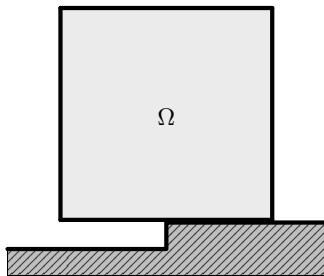
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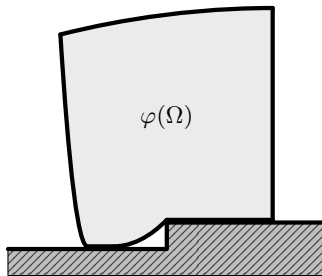
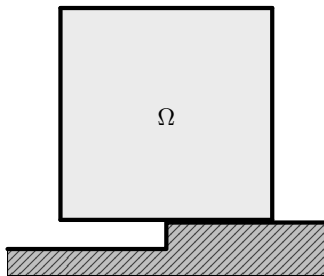
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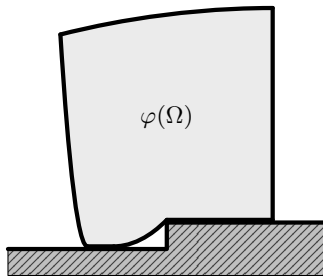
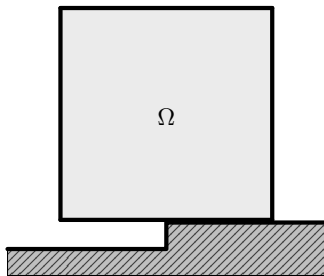
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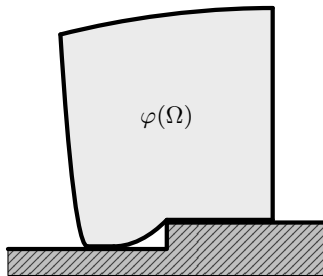
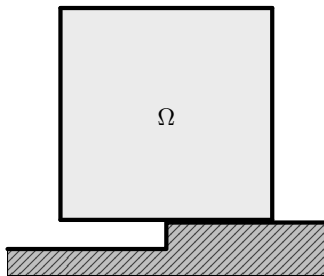
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- ▶ Complicated part: Contact Boundary.
- ▶ Rigid foundation can be described by gap-function  $g$  but we assume a flat foundation, i.e.  $g = 0$ .





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- ▶ For  $k = 1$  we get

$$\|u\|_{1,\Omega} = \left( \|u\|_{0,\Omega}^2 + \|\nabla u\|_{0,\Omega}^2 \right)^{1/2}, \quad u \in H^1(\Omega; \mathbb{R}^m).$$

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- ▶ Boundary traces of  $H^k$ -functions live in  $H^s$  spaces.

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- ▶ For  $s > 1/2$  we define the boundary space

$$H^{s-1/2}(\Gamma; \mathbb{R}^m) := \gamma_0(\mathcal{C}^\infty(\overline{\Omega}; \mathbb{R}^m))$$

and equip it with the norm

$$\|g\|_{s-\frac{1}{2}, \Gamma} := \inf \left\{ \|u\|_{s, \Omega} \mid u \in \mathcal{C}^\infty(\overline{\Omega}; \mathbb{R}^m), \gamma_0 u = g \right\}.$$



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- ▶ Same set for all  $s > 1/2$  but different norms.

# The Trace Theorem

## Theorem (Trace Theorem)

*Let  $1/2 < s \leq 1$ . Then the classical trace operator  $\gamma_0$  extends uniquely to a continuous surjective linear operator*

$$\gamma : H^s(\Omega; \mathbb{R}^m) \rightarrow H^{s-1/2}(\Gamma; \mathbb{R}^m).$$

*In particular, there exists a constant  $C_{\text{tr},s} > 0$  such that*

$$\|\gamma u\|_{s-\frac{1}{2},\Gamma} \leq C_{\text{tr},s} \|u\|_{s,\Omega} \quad \forall u \in H^s(\Omega; \mathbb{R}^m).$$

*In the special case  $s = 1$ , one has*

$$\ker(\gamma) = \overline{\mathcal{D}(\Omega; \mathbb{R}^m)}^{\|\cdot\|_{1,\Omega}} =: H_0^1(\Omega; \mathbb{R}^m).$$

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with norm

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- ▶ Moreover, we define

$$H_{00}^{s-1/2}(\Gamma_0; \mathbb{R}^m) := \left\{ g \in H^{s-1/2}(\Gamma_0; \mathbb{R}^m) \mid \text{Ext}_0(g) \in H^{s-1/2}(\Gamma; \mathbb{R}^m) \right\},$$

where  $\text{Ext}_0(g)$  denotes the zero extension of  $g$  from  $\Gamma_0$  to  $\Gamma$  with the norm

$$\|g\|_{s-\frac{1}{2}, 00, \Gamma_0} := \|\text{Ext}_0(g)\|_{s-\frac{1}{2}, \Gamma}.$$

# Dual Spaces of Boundary Spaces

► We define

$$H^{-1/2}(\Gamma; \mathbb{R}^m) := (H^{1/2}(\Gamma; \mathbb{R}^m))', \quad H^{-1/2}(\Gamma_0; \mathbb{R}^m) := (H_{00}^{1/2}(\Gamma_0; \mathbb{R}^m))'.$$

with duality pairings  $\langle \cdot, \cdot \rangle_\Gamma$  and  $\langle \cdot, \cdot \rangle_{\Gamma_0}$ , respectively.

## Theorem (Lifting Operator)

Let  $\Omega \subset \mathbb{R}^d$  be a bounded Lipschitz domain and let  $\Gamma_0 \subset \Gamma$  be relatively open.

- i) There exists a bounded linear operator  $\mathcal{L} : H^{1/2}(\Gamma; \mathbb{R}^d) \rightarrow H^1(\Omega; \mathbb{R}^d)$  such that

$$\gamma(\mathcal{L}g) = g \quad \forall g \in H^{1/2}(\Gamma; \mathbb{R}^d).$$

- ii) There exists a bounded linear operator  $\mathcal{L} : H_{00}^{1/2}(\Gamma_0; \mathbb{R}^d) \rightarrow H^1(\Omega; \mathbb{R}^d)$  such that for every  $g \in H_{00}^{1/2}(\Gamma_0; \mathbb{R}^d)$ ,

$$\gamma(\mathcal{L}g)|_{\Gamma_0} = g \quad \text{and} \quad \gamma(\mathcal{L}g) = 0 \text{ on } \Gamma \setminus \Gamma_0.$$

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# $H(\operatorname{div}; \Omega)$ for Matrix-Valued Fields

- ▶ The space  $\operatorname{Sym}^2(\mathbb{R}^d)$  of symmetric second order tensors is

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- ▶ On  $\mathbb{R}^{d \times d}$ , we use the Frobenius product

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- ▶ If  $\mathcal{A} : \Omega \rightarrow \mathcal{L}(\operatorname{Sym}^2(\mathbb{R}^d); \operatorname{Sym}^2(\mathbb{R}^d))$  is a fourth order tensor and  $s \in \operatorname{Sym}^2(\mathbb{R}^d)$ , we use the *double contraction*

$$(\mathcal{A}(x) : s)_{ij} := \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{kl}, \quad s : \mathcal{A}(x) : s := \sum_{i=1}^d \sum_{j=1}^d \sum_{k=1}^d \sum_{l=1}^d \mathcal{A}_{ijkl}(x) s_{ij} s_{kl}.$$

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- ▶ We define

$$H(\operatorname{div}; \Omega) := \{\tau \in L^2(\Omega; \operatorname{Sym}^2(\mathbb{R}^d)) \mid \operatorname{div} \tau \in L^2(\Omega; \mathbb{R}^d)\},$$

where  $\operatorname{div} \tau$  is understood row-wise in the weak sense.

# Normal Traces and Boundary Traction

- ▶ Let  $\mathbf{n} : \Gamma \rightarrow \mathbb{R}^d$  be the outward unit normal. For Lipschitz  $\Omega$  one has  $\mathbf{n} \in L^\infty(\Gamma; \mathbb{R}^d)$ .

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- ▶ If  $v \in H^1(\Omega; \mathbb{R}^d)$ , then its trace satisfies

$$\gamma v \in H^{1/2}(\Gamma; \mathbb{R}^d) \subset L^2(\Gamma; \mathbb{R}^d).$$

Hence the normal component

$$v_{\mathbf{n}} := (\gamma v) \cdot \mathbf{n}$$

lies in  $L^2(\Gamma)$ .



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Hence the normal component

$$v_{\mathbf{n}} := (\gamma v) \cdot \mathbf{n}$$

lies in  $L^2(\Gamma)$ .

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- ▶ On  $H(\operatorname{div}; \Omega)$ , the trace theorem does not hold. We need a different concept here.

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- ▶ In the smooth case,  $\gamma_n(\tau)$  coincides with the classical traction  $\tau \mathbf{n}$ . We thus write  $\tau \mathbf{n} = \gamma_n(\tau)$ .

# Normal Traces on Boundary Parts

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- ▶ Let  $\Gamma_0 \subset \Gamma$  be open in  $\Gamma$ .
- ▶ The normal trace on  $\Gamma_0$  is defined as the unique element  $\gamma_{n,0}(\tau) \in H^{-1/2}(\Gamma_0; \mathbb{R}^d)$  given by

$$\langle \gamma_{n,0}(\tau), g \rangle_{\Gamma_0} := \langle \gamma_n(\tau), \text{Ext}_0(g) \rangle_{\Gamma}, \quad g \in H_{00}^{1/2}(\Gamma_0; \mathbb{R}^d),$$

where  $\text{Ext}_0(g)$  denotes the zero extension of  $g$  from  $\Gamma_0$  to  $\Gamma$ .

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1 Introduction

2 Sobolev Spaces

3 Signorini's Problem

- Geometric Setup
- Weak Formulations
- Strong Formulations
- Mixed Formulations

4 Further Topics

- ▶ Elastic Body  $\Omega \subset \mathbb{R}^d$ ,  $d \in \{2, 3\}$  open and bounded domain with Lipschitz boundary  $\Gamma := \partial\Omega$ .



# Boundary Decomposition

- ▶ Elastic Body  $\Omega \subset \mathbb{R}^d$ ,  $d \in \{2, 3\}$  open and bounded domain with Lipschitz boundary  $\Gamma := \partial\Omega$ .
- ▶ Decompose  $\Gamma$  into three relatively open and pairwise disjoint parts

$$\Gamma = \overline{\Gamma_D} \dot{\cup} \overline{\Gamma_N} \dot{\cup} \overline{\Gamma_C}, \quad |\Gamma_D|, |\Gamma_C| > 0.$$

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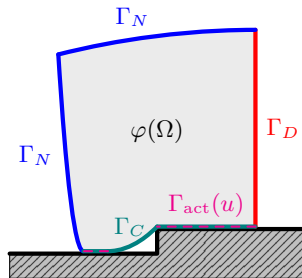
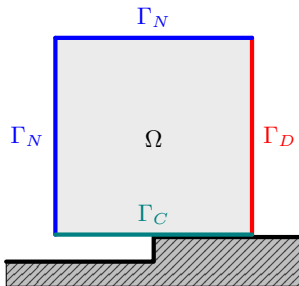
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- ▶ The body in equilibrium  $\Omega$  configuration is  $\varphi(\Omega)$  with  $\varphi(x) = x + u(x)$ .

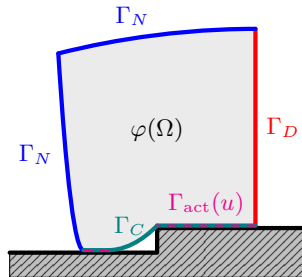
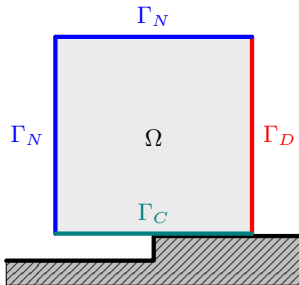
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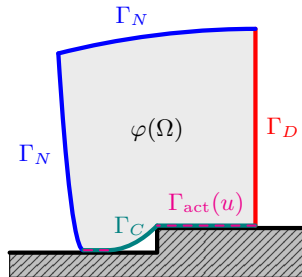
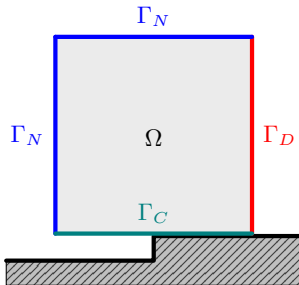
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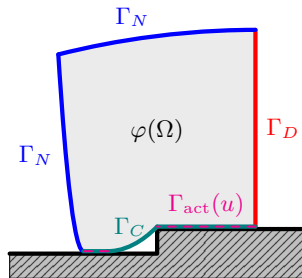
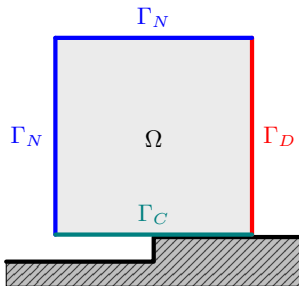
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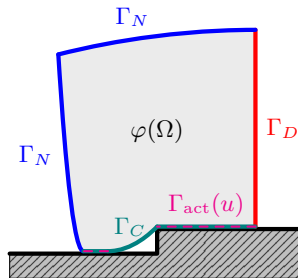
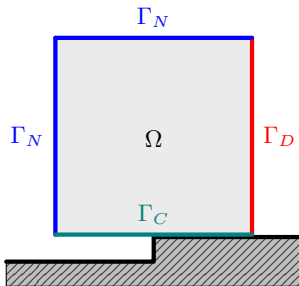
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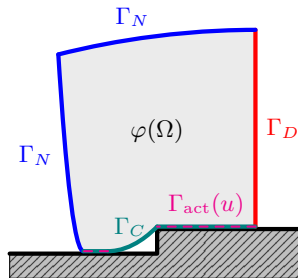
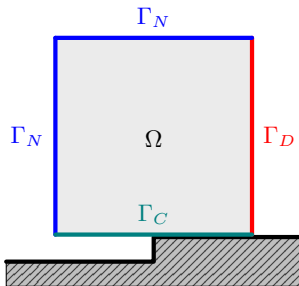
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## 4 Further Topics

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- ▶ Homogeneous Dirichlet boundary conditions are incorporated in the underlying Hilbert space

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# Small Strain Elasticity

- ▶ *Small strain tensor:*

$$\varepsilon : H^1(\Omega; \mathbb{R}^d) \rightarrow L^2(\Omega; \text{Sym}^2(\mathbb{R}^d)), \quad u \mapsto \varepsilon(u) := \frac{1}{2} (\nabla u + \nabla u^\top).$$

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- ▶ *(Cauchy) stress tensor:*

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# Virtual Work: External and Internal Forces

- ▶ Let  $f \in L^2(\Omega; \mathbb{R}^d)$  be given *volume force density* and  $F \in L^2(\Gamma_N; \mathbb{R}^d)$  describe the *surface load* which we impose on  $\Gamma_N$ .

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$$a(u, v) := \int_{\Omega} \sigma(u) : \varepsilon(v) \, dx, \quad u, v \in V.$$

# Standard Properties of $a$ and $\ell$

## Proposition

The symmetric bilinear form  $a : V \times V \rightarrow \mathbb{R}$  defined above is

i) bounded, i.e. for all  $v, w \in V$  there exists  $C_1 < \infty$  such that

$$|a(u, v)| \leq C_1 \|u\|_{1,\Omega} \|v\|_{1,\Omega} \quad \text{with} \quad C_1 = C_A,$$

ii)  $V$ -elliptic, i.e. for all  $v \in V \setminus \{0\}$  there exist  $C_2 > 0$  such that

$$a(v, v) \geq C_2 \|v\|_{1,\Omega}^2 \quad \text{with} \quad C_2 = \frac{\alpha_A}{C_K^2}.$$

Moreover, the linear functional  $\ell$  defined above is bounded, i.e. for all  $v \in V$  there exists  $C_3 > 0$  such that

$$|\ell(v)| \leq C_3 \|v\|_{1,\Omega} \quad \text{with} \quad C_3 = \max\{1, C_{\text{tr}}\} (\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N})$$

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## Problem (Signorini's Problem – Energy Minimization)

Find  $u \in K$  such that

$$\mathcal{J}(u) = \min_{v \in K} \mathcal{J}(v) = \min_{v \in K} \frac{1}{2} a(v, v) - \ell(v).$$

## Lemma (Weak lower semicontinuity of the norm)

*Let  $(X, \|\cdot\|_X)$  be a normed space. Then  $\|\cdot\|_X : X \rightarrow \mathbb{R}_{\geq 0}$  is a weakly lower semicontinuous functional, i.e. if  $x_j \rightharpoonup x$  in  $X$ , then*

$$\|x\|_X \leq \liminf_{j \rightarrow \infty} \|x_j\|_X.$$

**Proof:** Omitted here.

# Existence of Minimizer

## Lemma ( $\mathcal{J}$ is coercive and w.l.s.c.)

The functional  $\mathcal{J}$  is

i) coercive on  $V$ , i.e.

$$\mathcal{J}(v) \rightarrow \infty \quad \text{as} \quad \|v\|_{1,\Omega} \rightarrow \infty,$$

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► By boundedness of  $\ell$  we have  $|\ell(v)| \leq \|\ell\|_{V'} \|v\|_{1,\Omega}$ . Hence

$$\mathcal{J}(v) = \frac{1}{2} a(v, v) - \ell(v) \geq \frac{\alpha}{2} \|v\|_{1,\Omega}^2 - \|\ell\|_{V'} \|v\|_{1,\Omega} \xrightarrow{\|v\|_{1,\Omega} \rightarrow \infty} \infty.$$

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► This proves i).

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- ▶ Therefore,

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$$\mathcal{J}(u) \leq \liminf_{k \rightarrow \infty} \mathcal{J}(v_{j_k}) = M = \inf_{v \in K} \mathcal{J}(v).$$

□



## Proposition (Uniqueness of a Minimizer of $\mathcal{J}$ )

*If there exists a minimizer of the functional  $\mathcal{J}$ , it is unique.*

**Proof:**  $\mathcal{J}$  is strictly convex on  $V$  and thus on the convex set  $K$ . Thus, a minimizer  $u \in K$  is unique.  $\square$

# A-Priori Stability Bound

## Proposition (Stability estimate for the minimizer)

*The Unique minimizer  $u \in K$  of  $\mathcal{J}$  satisfies the a-priori bound*

$$\|u\|_{1,\Omega} \leq \frac{2 C_K^2 \max\{1, C_{\text{tr}}\}}{\alpha_{\mathcal{A}}} (\|f\|_{0,\Omega} + \|F\|_{0,\Gamma_N}).$$

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- ▶ Finally,

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# Variational Inequality as an Optimality Condition

- Fréchet Derivative of  $\mathcal{J}$  at  $u \in V$  in direction  $h \in V$  is  $\mathcal{J}'(u)[h] = a(u, h) - \ell(h)$ .

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- Fréchet Derivative of  $\mathcal{J}$  at  $u \in V$  in direction  $h \in V$  is  $\mathcal{J}'(u)[h] = a(u, h) - \ell(h)$ .

## Proposition (First-order optimality condition on convex sets)

*A function  $u \in K$  minimizes  $\mathcal{J}$  over  $K$  if and only if it satisfies the variational inequality*

$$\mathcal{J}'(u)[v - u] \geq 0 \quad \forall v \in K.$$



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